

WaterGS: Physically-Based Imaging in Gaussian Splatting for Underwater Scene Reconstruction

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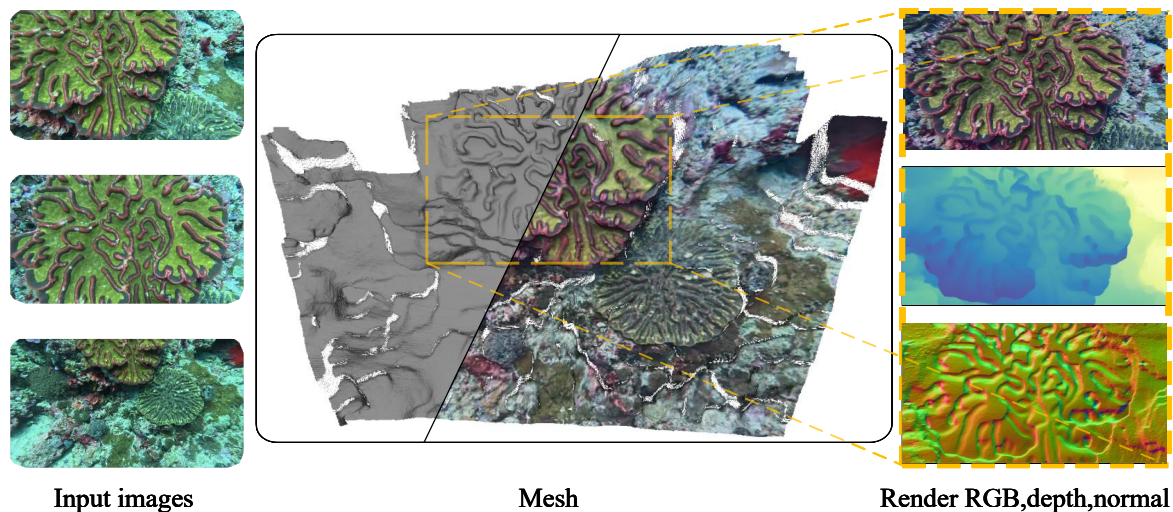


Figure 1: Our Method overcomes underwater image degradation (sample inputs, left) to jointly reconstruct high-fidelity 3D mesh geometry (center left) and recover clear, de-watered surface appearance (center right, highlighted coral). The rich representation also allows for rendering of clear RGB, depth, and normal maps (right detail).

Abstract

Reconstructing underwater object geometry from multi-view images is a long-standing challenge in computer graphics, primarily due to image degradation caused by underwater scattering, blur, and color shift. These degradations severely impair feature extraction and multi-view consistency. Existing methods typically rely on pre-trained image enhancement models as a preprocessing step, but often struggle with robustness under varying water conditions. To overcome these limitations, we propose WaterGS, a novel framework for underwater surface reconstruction that jointly recovers accurate 3D geometry and restores true object colors. The core of our approach lies in introducing a Physically-Based imaging model into the rendering process of 2D Gaussian Splatting. This enables accurate separation of true object colors from water-induced distortions, thereby facilitating more robust photometric alignment and denser geometric reconstruction across views. Building upon this improved photometric consistency, we further introduce a Gaussian bundle adjustment scheme guided by our physical model to jointly optimize camera poses and geometry, enhancing reconstruction accuracy. Extensive experiments on synthetic and real-world datasets show that WaterGS achieves robust, high-fidelity reconstruction directly from raw underwater images, outperforming prior approaches in both geometric accuracy and visual consistency.

CCS Concepts

• **Computing methodologies** → **Reconstruction; Image-based rendering;**

1. Introduction

Underwater 3D reconstruction is essential for many scientific and engineering tasks, such as coral reef monitoring [AL20], underwater archaeology [JRBF*17], and infrastructure inspection [LZL*24], where high-fidelity geometric models are critical for detailed analysis, measurement, and preservation. However, despite substantial progress in computer graphics and vision, underwater reconstruction remains fundamentally challenging due to its unique imaging conditions. These challenges prevent the direct application of existing methods, which often leads to degraded reconstruction quality and inconsistent results.

The core difficulty in underwater reconstruction lies in the inherently complex nature of underwater imaging. Light propagation is strongly affected by wavelength-dependent absorption, which causes noticeable color distortion and rapid signal loss over distance [HLY*22, AT18, McG80]. At the same time, suspended particles in the water scatter light, leading to image blur, lower contrast, and loss of fine scene details. These effects are further made worse by non-uniform lighting conditions, which introduce additional ambiguity and inconsistency across views. As a result, the photometric consistency assumption, which is crucial to multi-view methods such as structure-from-motion (SfM) and multi-view stereo (MVS), often fails to hold. In addition, because of the difference in refractive indices between water and camera lenses, light bends at the water-glass interface, introducing systematic geometric errors. This refraction makes the conventional pinhole camera model less reliable for underwater reconstruction [SK12, Jor14, SSNK24].

To cope with these challenges, traditional approaches typically adopt a staged pipeline that first applies image enhancement techniques [AADVB17], followed by refractive calibration and standard SfM/MVS algorithms [Jor14, SSNK24]. However, this sequential strategy often suffers from error accumulation, where artifacts introduced during enhancement negatively impact feature matching and camera pose estimation in subsequent stages. Recently, neural radiance fields (NeRF) [MST*21] have shown remarkable success in modeling complex appearance and have sparked new insights into underwater scene reconstruction. Building upon this direction, SeaThru-NeRF [LPP*23] shares a similar motivation with our method by explicitly integrating an underwater imaging model into the NeRF framework, achieving high-quality novel view synthesis under challenging underwater conditions. However, this implicit volumetric representation inherently lacks strong geometric constraints, making it difficult to extract accurate and detailed surface geometry. Subsequent approach based on neural implicit surfaces (neural SDFs [WLL*21, YGKL21]), e.g., UW-SDF [CTW*24], attempts to bridge this gap by leveraging Eikonal loss to enforce surface smoothness and regularity. While it helps mitigate surface ambiguity, this method relies heavily on pre-trained image enhancement modules and geometric prior estimations. Such strong dependencies significantly compromise robustness and often lead to unstable reconstruction results under varying underwater conditions.

The recent emergence of point-like implicit representations has presented a promising avenue toward efficient 3D scene reconstruction. Among these, 3D Gaussian Splatting (3DGS) [KKLD23]

and its 2D extension (2DGS) [HYC*24] have demonstrated strong capabilities in capturing geometric details while maintaining real-time rendering performance. Among existing methods, SeaSplat [YLG24] stands out as the most relevant to our work, as it incorporates an underwater imaging model into the 3DGS rendering process. However, despite the effectiveness in appearance restoration, SeaSplat suffers from noisy reconstructions due to floating Gaussians. Moreover, due to the limited representational precision of 3D Gaussians, the method often struggles to accurately reconstruct complex structures, e.g., underwater reefs and corals.

To address these challenges, we propose WaterGS, a novel underwater reconstruction framework built upon 2DGS. Compared to 3DGS, 2DGS offers improved robustness and finer geometric details, making it particularly well-suited for handling the complexities of underwater scenes. Our pipeline begins with leveraging a state-of-the-art SfM method, VGGT [WCK*25], to obtain initial dense point clouds and camera parameters. The dense point clouds provide a stronger prior for 2DGS initialization. We further introduce a physically-based underwater imaging model, which represents image degradation with spatially varying absorption and scattering, enabling direct optimization of the underlying scene color. This physically-based modeling substantially enhances multi-view consistency and pixel-wise feature matching, facilitating more accurate surface reconstruction. Additionally, we introduce a Gaussian bundle adjustment module that jointly optimizes scene geometry and camera extrinsics in an end-to-end manner. While prior methods rely solely on initial camera poses estimated from raw underwater images, our joint optimization effectively corrects pose inaccuracies during training, leading to more detailed reconstructions without compromising computational efficiency. Our contributions are summarized as follows:

- A novel reconstruction framework that integrates a learnable physically-based underwater imaging model into 2D Gaussian Splatting, enabling joint recovery of accurate geometry and clear appearance.
- A Gaussian bundle adjustment strategy that leverages physically-corrected photometric cues to jointly optimize camera poses and scene parameters, enhancing geometric fidelity.
- Comprehensive experiments on diverse synthetic and real-world datasets demonstrate that our method outperforms state-of-the-art approaches in both geometric accuracy and visual fidelity.

2. Related Work

2.1. Traditional Underwater 3D Reconstruction

Traditional pipelines adapt Structure-from-Motion (SfM) and Multi-View Stereo (MVS) techniques for underwater environments through specialized, sequential processing [HWS*23]. These methods are applied across marine biology, underwater archaeology, and habitat mapping, but face fundamental challenges in aquatic environments.

To address geometric distortions from camera housing refraction, researchers have developed approaches including explicit refractive interface modeling [Jor14, FQC*21], adjusted camera intrinsic parameters to approximate refractive effects [SK12], and refraction-aware SfM algorithms that integrate refractive geometry

into pose estimation [HLP23]. Complementing these geometric solutions, underwater image enhancement algorithms such as PUIE-Net [FWH*22] and FA+ [JYC*23], along with other physics-based restoration models [BJRPW16, AL20, ZJZ*23], serve as pre-processing steps to counteract light absorption and scattering effects. The foundational Jaffe-McGlamery model [Jaf90, McG80] decomposes underwater image formation into direct transmission, forward scattering, and backscattering components, providing the physical basis for many enhancement approaches.

For extremely challenging conditions, active sensing technologies including sonar, laser line scanning [LZL*24], photometric stereo [FQC*21], and structured light systems provide alternative 3D information sources where passive optical methods struggle.

Despite these advances, traditional methods suffer from critical limitations. The sequential "enhance-then-reconstruct" paradigm leads to error propagation, where pre-processing artifacts corrupt geometric inference and photometric consistency [SK12, HWS*23]. Traditional bundle adjustment relies purely on geometric reprojection errors and lacks understanding of underwater imaging physics, making it particularly vulnerable when water effects violate photometric assumptions. Underwater conditions compromise initial pose estimation due to poor feature quality and unmodeled refractive distortions, creating unreliable starting points for reconstruction. While recent neural bundle adjustment methods like BARF [LMTL21] demonstrate joint pose-scene optimization, they assume clear imaging conditions incompatible with underwater environments.

2.2. Neural Rendering for Underwater Scenes

Neural Radiance Fields (NeRF) [MST*21] have revolutionized 3D scene representation through continuous volumetric modeling. Underwater adaptations such as SeaThru-NeRF [LPP*23] explicitly incorporate underwater light propagation models into the NeRF rendering equation, enabling joint water parameter estimation and clear appearance recovery. WaterNeRF [SRS23] combines physical models with data-driven techniques to dynamically model light attenuation and scattering effects, while Zhang et al. [ZJR23] integrated underwater illumination effects into differentiable rendering frameworks.

While these methods demonstrate impressive novel view synthesis and appearance restoration, their primary optimization target is rendering quality rather than explicit surface reconstruction. UW-SDF [CTW*24] leverages depth and normal estimation networks to constrain implicit surface optimization, but extracting explicit triangle meshes from volumetric representations often introduces noise or excessive smoothing.

3D Gaussian Splatting (3DGS) [KKLD23] offers an alternative through explicit primitive-based representation, achieving real-time, high-quality rendering via differentiable rasterization. UW-GS [WAZB25] focuses on distractor-aware reconstruction using color appearance models and physics-based density control for dynamic underwater elements. SeaSplat [YLG24] also integrates underwater imaging models with 3DGS for effective appearance restoration. However, robust extraction of high-fidelity surface geometry from volumetric 3D Gaussians remains challenging since

the primitives are fundamentally three-dimensional and do not directly correspond to surface elements. To address these limitations, 2D Gaussian Splatting [HYC*24] employs oriented 2D Gaussian discs that directly represent scene surfaces, providing inherently surface-aligned geometry suitable for high-quality surface reconstruction.

These concurrent developments in underwater 3DGS typically employ approaches where physical effects are modeled through separate networks or post-processing stages. In contrast, our WaterGS framework distinguishes itself by directly integrating a learnable physical imaging model into the core 2D Gaussian Splatting rendering process, enabling seamless joint optimization of geometry and appearance. Furthermore, while existing methods generally assume accurate initial camera poses from SfM, WaterGS addresses the fundamental challenge of pose inaccuracy in underwater conditions through its Gaussian bundle adjustment, reducing dependency on potentially unreliable traditional pipelines.

3. Preliminaries

3.1. 2D Gaussian Splatting

We employ 2D Gaussian Splatting (2DGS) as an explicit, surface-aligned scene representation. Unlike 3D Gaussian Splatting, which uses volumetric 3D ellipsoids, 2DGS represents a scene as a set of 2D Gaussian primitives (surfels) that lie on the surface.

Each surfel g_k has learnable parameters: its 3D position $\mathbf{p}_k \in \mathbb{R}^3$; an orthonormal local tangent frame with $\mathbf{u}_k, \mathbf{v}_k \in \mathbb{R}^3$ and surface normal $\mathbf{n}_k = \mathbf{u}_k \times \mathbf{v}_k$; anisotropic scales $s_{u,k}, s_{v,k} \in \mathbb{R}_+$ controlling Gaussian extent along the tangent axes; opacity $\alpha_k \in [0, 1]$; and a view-dependent appearance $\mathbf{c}_k(\cdot) \in \mathbb{R}^3$ (e.g., parameterized by Spherical Harmonics coefficients).

During rendering, the 3D surfels are projected to the image plane and accumulated via differentiable rasterization to synthesize pixel colors.

3.2. Underwater Imaging Model

Building on 2DGS, we incorporate underwater light-propagation physics via the Jaffe-McGlamery model [McG80, Jaf90]. The observed radiance at pixel x is

$$\mathbf{I}_{\text{obs}}(x) = \mathbf{J}(x) \odot \mathbf{T}(x) + \mathbf{B}(x) \odot (\mathbf{1} - \mathbf{T}(x)), \quad (1)$$

where $\mathbf{J}(x), \mathbf{B}(x) \in \mathbb{R}^3$ are the clear scene radiance and ambient backscatter color, respectively, and $\mathbf{T}(x) \in \mathbb{R}^3$ is the (per-channel) transmission. Transmission follows exponential decay with camera-to-surface distance $d(x)$:

$$\mathbf{T}(x) = \exp(-\beta d(x)), \quad (2)$$

with $\beta \in \mathbb{R}^3$ the wavelength-dependent attenuation coefficients; the exponential is applied element-wise, and $\odot$ denotes element-wise multiplication. (When rendered depth z exists, set $d(x) \leftarrow z(x)$.)

The conventional model assumes spatially constant water parameters, i.e., β and $\mathbf{B}$ do not vary with x . In practice, this fails to capture local changes in water properties. In Sec. 4.2 we address this by integrating a learnable, spatially varying extension into the 2DGS rendering pipeline.

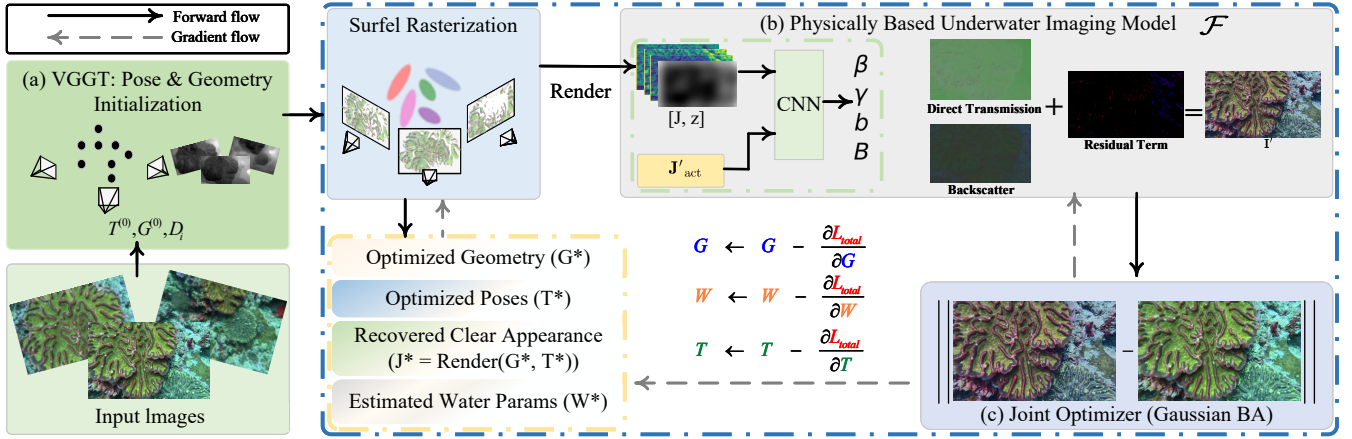


Figure 2: Overview of WaterGS. From unposed underwater images, VGGT provides robust initialization—camera poses $\{T_i^{(0)}\}_{i=1}^N$, geometry $G^{(0)}$, and depth maps $\{D_i\}_{i=1}^N$. (1) **Surfel Rasterization** (2DGS) renders per-view clear appearance and depth $[J, z]$. (2) A **Physically Based Underwater Imaging Model** $\mathcal{F}$ takes $[J, z]$; a lightweight CNN predicts per-pixel water parameters $\mathbf{W} = \{\beta, \gamma, b, \mathbf{B}\}$, and a learnable global ambient-light color vector $\mathbf{J}_{\text{prime}}$ (activated to $\mathbf{J}'_{\text{act}}$) to compose Direct Transmission + Backscatter + Residual and synthesize $\mathbf{I}'$. (3) **Gaussian Bundle Adjustment** jointly optimizes G , T , and $\mathbf{W}$ under the total loss $\mathcal{L}_{\text{total}}$, yielding G^* , T^* , $\mathbf{J}^*$, and $\mathbf{W}^*$ for high-fidelity 3D reconstruction and appearance restoration. Notation: β, γ, b are per-pixel scalars; $\mathbf{B}, \mathbf{J}, \mathbf{I}'$ are RGB vectors; z is a scalar.

4. Method

Given a set of N underwater RGB images $\{\mathbf{I}_i\}_{i=1}^N$, our objective is to reconstruct an explicit 3D geometric representation G and simultaneously recover its intrinsic, clear surface appearance $\mathbf{J}$. We formulate underwater reconstruction as a Physically-Based inverse-rendering problem built upon 2D Gaussian Splatting (2DGS). As outlined in Fig. 2, *WaterGS* combines robust initialization (Sec. 4.1), a learnable Physically-Based underwater imaging model within the rendering pipeline (Sec. 4.2), and a Gaussian bundle adjustment strategy (Sec. 4.3). This enables joint optimization of explicit scene geometry, clear appearance, camera poses, and water parameters using a unified set of loss functions (Sec. 4.4), providing a comprehensive solution directly from raw underwater imagery.

4.1. Robust Geometric Initialization via VGGT

Given unposed images $\{\mathbf{I}_i\}_{i=1}^N$, VGGT [WCK*25] predicts in a single forward pass a consistent set of initial camera poses $\{T_i^{(0)}\}_{i=1}^N$, with $T_i^{(0)} \in SE(3)$, alongside per-image depth maps $\{D_i\}_{i=1}^N$ and an initial 3D point cloud $P^{(0)}$. This end-to-end initialization mitigates the error accumulation common in staged SfM and is more robust to underwater image degradation. We initialize our 2D Gaussian surfels $G^{(0)}$ by converting each point in $P^{(0)}$ into a surfel whose initial color is sampled from the corresponding input views. We use the VGGT depth maps as a geometric prior through the depth-consistency term $\mathcal{L}_{\text{depth}}$ (Sec. 4.4), which helps stabilize geometry in weakly textured underwater regions. In contrast, general-purpose monocular depth estimators such as DepthAnything [YKH*24] and Marigold [KOH*24] tend to perform poorly on underwater scenes and offer less reliable guidance.

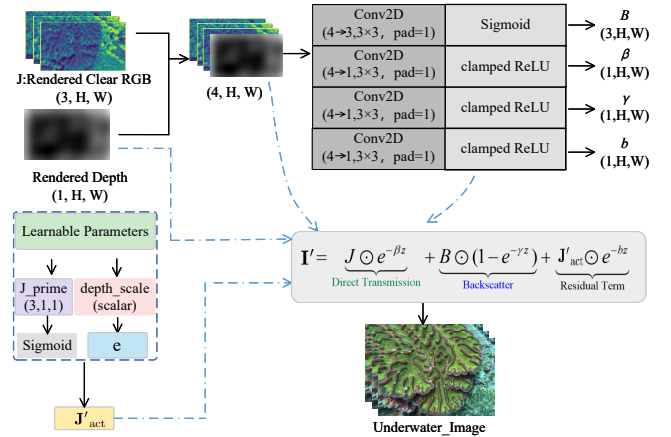


Figure 3: Physically Based Underwater Imaging Model. Given rendered clear appearance and depth $[J, z]$, a CNN predicts per-pixel water parameters $\{\beta, \gamma, b\}$ and backscatter color $\mathbf{B}$. A learnable global ambient light $\mathbf{J}_{\text{prime}}$ is activated to $\mathbf{J}'_{\text{act}}$, and the internal depth rescale ($z \leftarrow e^{\text{depth_scale}} z$) is applied in the formation (Eq. 3).

4.2. Physically-Based Underwater Rendering and Appearance Recovery

To accurately model scenes and recover their clear, intrinsic appearance, we introduce a unified rendering and appearance recovery module. At its core is our physically-based underwater imaging model (Fig. 3), integrated into the differentiable 2DGS rasterization process. This model simulates underwater image formation and provides physics-based constraints that guide optimization toward visually faithful, plausible results.

For each pixel, the 2DGS rasterizer provides the clear surface color $\mathbf{J} \in \mathbb{R}^3$ and a scalar depth $z \in \mathbb{R}$. During training, to absorb global scale/illumination ambiguity, we introduce a learnable scalar depth_scale and internally rescale depth as $z \leftarrow \exp(\text{depth_scale})z$; for notational simplicity, we continue to write z . The learnable imaging model predicts the observed underwater color $\mathbf{I}' \in \mathbb{R}^3$ as

$$\mathbf{I}' = \underbrace{\mathbf{J} \odot e^{-\beta z}}_{\text{Direct Transmission}} + \underbrace{\mathbf{B} \odot (1 - e^{-\gamma z})}_{\text{Backscatter}} + \underbrace{\mathbf{J}'_{\text{act}} \odot e^{-bz}}_{\text{Residual Term}}, \quad (3)$$

where $\odot$ denotes element-wise multiplication, $\beta, \gamma, b \in \mathbb{R}_{\geq 0}$ are per-pixel scalar attenuations, and $\mathbf{B} \in \mathbb{R}^3$ is a per-pixel RGB backscatter color. Unlike fixed-parameter models, the water parameters $\mathbf{W} = \{\beta, \gamma, b, \mathbf{B}\}$ are predicted per pixel by a lightweight CNN. The network takes the 4-channel input $[\mathbf{J}, z]$ and uses four parallel convolutional branches to output the parameter maps, enabling adaptation to localized water variations. This per-pixel formulation naturally accommodates inhomogeneous water and local illumination (e.g., near lamps or particle clouds), where turbidity and backscatter can vary across the image plane. Branch heads enforce valid ranges: we constrain β, γ, b to be nonnegative (e.g., via Softplus or clamped ReLU) and map $\mathbf{B}$ to $[0, 1]^3$ (e.g., via Sigmoid or clamp).

Unlike prior underwater models that fold ambient illumination into the backscatter color, we explicitly decouple it via a residual branch parameterized by a learnable global ambient light vector $\mathbf{J}_{\text{prime}} \in \mathbb{R}^3$. We obtain the active term by an element-wise logistic sigmoid, $\mathbf{J}'_{\text{act}} = \text{sigmoid}(\mathbf{J}_{\text{prime}}) \in [0, 1]^3$. This separation prevents $\mathbf{B}$ from absorbing scene-wide color casts and stabilizes depth-scale estimation, yielding more robust optimization across diverse water conditions.

Resolving Appearance–Parameter Ambiguity. Optimizing $\mathbf{J}$ and $\mathbf{W}$ with a photometric loss *alone* is ill-posed: different $(\mathbf{J}, \mathbf{W})$ pairs can yield the same observation $\mathbf{I}'$, which can corrupt the recovered appearance. We address this with a feedback guidance loss based on Retinex theory [KES*03]. We estimate a stable reflectance by decomposing the observed image into reflectance R and illumination L via the variational problem

$$(R^*, L^*) = \arg \min_{R, L} \|\mathbf{I} - R \odot L\|_2^2 + \lambda_R \phi_R(R) + \lambda_L \phi_L(L), \quad (4)$$

where $\odot$ denotes element-wise multiplication, the data term enforces reconstruction fidelity, and ϕ_R, ϕ_L are standard priors (TV for R , smoothness for L). We take the solution's reflectance R^* as our physically grounded target and, for brevity, write $\mathcal{R}(\mathbf{I}) := R^*$. The feedback loss is then $\mathcal{L}_{\text{feedback}} = \|\mathcal{R}(\mathbf{I}) - \mathbf{J}\|_1$ (see Sec. 4.4).

4.3. Gaussian Bundle Adjustment and Joint Optimization

Traditional bundle adjustment based only on geometric reprojection is brittle underwater. Instead, *WaterGS* performs joint optimization by combining robust VGGT initialization with a fully differentiable 2DGS pipeline. We seek geometry G , camera poses T , and imaging-network weights θ_W that best explain the observations:

$$G^*, T^*, \theta_W^* = \arg \min_{G, T, \theta_W} \mathcal{L}_{\text{total}}(G, T, \theta_W), \quad (5)$$

where $\mathcal{L}_{\text{total}}$ (Sec. 4.4) combines photometric, geometric, guidance, and regularization terms. For view i , the 2DGS rasterizer produces

$[\mathbf{J}_i, z_i] = \mathcal{G}(G, T_i)$, the parameter-predicting CNN yields per-pixel water parameters

$$\mathbf{W}_i = \Phi([\mathbf{J}_i, z_i]; \theta_W), \quad \text{with } \mathbf{W}_i = \{\beta_i, \gamma_i, b_i, \mathbf{B}_i\}, \quad (6)$$

and the imaging model synthesizes $\mathbf{I}'_i = \mathcal{F}(\mathbf{J}_i, z_i, \mathbf{W}_i)$, where $\mathcal{F}$ implements the physical formation in Eq. (3). Here $\mathbf{J}_i \in \mathbb{R}^3$ is the per-pixel clear color, $z_i \in \mathbb{R}$ is the scalar depth, and $\mathbf{I}'_i \in \mathbb{R}^3$ is the synthesized underwater color. Each pose $T_i \in SE(3)$, parameterized by θ_{T_i} , transforms each surfel's world-space parameters (center $\mathbf{p}_k$, tangent frame $\mathbf{u}_k, \mathbf{v}_k$, scales $s_{u,k}, s_{v,k}$, opacity α_k , and view-dependent appearance), which $\mathcal{G}$ then projects and accumulates to form $[\mathbf{J}_i, z_i]$.

Because the pipeline is differentiable, gradients propagate to pose via the chain rule:

$$\nabla_{\theta_{T_i}} \mathcal{L}_{\text{total}} = \frac{\partial \mathcal{L}_{\text{total}}}{\partial \mathbf{I}'_i} \frac{\partial \mathbf{I}'_i}{\partial [\mathbf{J}_i, z_i]} \frac{\partial [\mathbf{J}_i, z_i]}{\partial \theta_{T_i}}, \quad (7)$$

with analogous paths for θ_G and θ_W handled by backpropagation. Jointly updating $(\theta_{T_i}, \theta_G, \theta_W)$ reduces pose drift and achieves globally consistent reconstructions under the physics-aware image formation.

4.4. Loss Functions

The optimization of *WaterGS* is driven by a composite loss $\mathcal{L}_{\text{total}}$ that combines a primary photometric data term with guidance and regularization losses. Unless noted, pixelwise terms are averaged over all image pixels and views.

Photometric Loss. The primary supervisory signal is the photometric loss $\mathcal{L}_{\text{photo}}$, which encourages the predicted images $\mathbf{I}'_{ij}$ to match the observations $\mathbf{I}_{ij}$. It is a weighted sum of an ℓ_1 term and a structural similarity (SSIM) term [WBSS04]:

$$\mathcal{L}_{\text{photo}} = (1 - \lambda_{\text{ssim}}) \sum_{i,j} \|\mathbf{I}'_{ij} - \mathbf{I}_{ij}\|_1 + \lambda_{\text{ssim}} \sum_{i,j} (1 - \mathcal{L}_{\text{SSIM}}(\mathbf{I}'_{ij}, \mathbf{I}_{ij})). \quad (8)$$

Guidance and Regularization. To guide optimization and ensure plausible results, we incorporate a depth-consistency term and a Retinex-based feedback term. With VGGT priors (Sec. 4.1), the *depth consistency* is

$$\mathcal{L}_{\text{depth}} = \frac{1}{N_{\text{pix}}} \sum_{i,j} |z_{ij} - (D_i)_j|, \quad (9)$$

where N_{pix} is the total number of pixels across all views. To resolve appearance ambiguity (Sec. 4.2), we compute a stable reflectance R^* per image via Eq. (4) and use the shorthand $\mathcal{R}(\mathbf{I}_i) := R_i^*$. The *Retinex feedback* is

$$\mathcal{L}_{\text{feedback}} = \frac{1}{N_{\text{pix}}} \sum_{i,j} \|\mathcal{R}(\mathbf{I}_i)_j - \mathbf{J}_{ij}\|_1, \quad (10)$$

which encourages the rendered clear appearance $\mathbf{J}$ to align with the stable reflectance. Since branch activations already constrain $\beta, \gamma, b \geq 0$ and $\mathbf{B} \in [0, 1]^3$ (Sec. 4.2), we do not include a separate non-negativity penalty. Finally, to prevent overfitting in the parameter-predicting CNN, we apply standard ℓ_2 *weight regularization*, $\mathcal{L}_{\text{reg}} = \|\mathbf{W}_{\text{net}}\|_F^2$, where $\mathbf{W}_{\text{net}}$ includes all CNN parameters.

Total Loss. The final objective for joint optimization is a weighted sum of all components:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{photo}} \mathcal{L}_{\text{photo}} + \lambda_{\text{feedback}} \mathcal{L}_{\text{feedback}} + \lambda_{\text{depth}} \mathcal{L}_{\text{depth}} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}}, \quad (11)$$

with fixed positive weights $\lambda_{\text{photo}}, \lambda_{\text{feedback}}, \lambda_{\text{depth}}, \lambda_{\text{reg}}$. This objective is minimized over geometry G , poses T , and imaging-network weights θ_W (Sec. 4.3).

5. Experiments

We conducted comprehensive experiments to evaluate WaterGS on both geometric reconstruction and appearance restoration across diverse synthetic and real-world underwater scenarios. For further ablations and comparisons, see the supplementary material.

5.1. Datasets

We utilize four datasets spanning controlled synthetic environments to challenging in-the-wild scenarios. For geometric evaluation with ground truth, we employ DTU-Water, a synthetic benchmark generated by applying realistic underwater degradation effects from UWNR [YCL*22] to the classic DTU dataset [AJV*16], and UW-Bundle [SIJR17], a real-world dataset captured in a controlled laboratory pool that uniquely provides high-precision ground-truth meshes and corresponding in-air reference images.

For appearance restoration and multi-view consistency evaluation without ground-truth geometry, we use DRUVA [VKR23], featuring video sequences of underwater artifacts captured under challenging lighting conditions, and Coral, our collected dataset comprising underwater coral reef videos captured with consumer-grade cameras in unconstrained environments.

5.2. Evaluation Metrics

We assess WaterGS performance across three key dimensions. For **geometric accuracy**, we report the Mean Chamfer Distance (CD) [FSG17] between our reconstructed meshes and the ground truth. For **image restoration quality** on in-the-wild scenes, we employ the no-reference NIQE [MSB12] and UCIQE [YS15] metrics.

Finally, to quantify **multi-view consistency**—a critical advantage over single-frame enhancement methods—we introduce a pairwise evaluation. After rendering the restored appearance sequence from the optimized trajectory, we compute the Pairwise LPIPS and Pairwise Hue Correlation between adjacent frames. This specific application of LPIPS is designed to measure perceptual stability from one view to the next, directly testing the temporal coherence that single-frame methods cannot guarantee.

5.3. Implementation and Baselines

Implementation Details. Our framework is implemented using the 2DGS codebase [HYC*24] and trained for 15,000 iterations on an NVIDIA RTX 3090 GPU. Loss function weights (Equation (11)) are set to $\lambda_{\text{photo}} = 1.0$, $\lambda_{\text{feedback}} = 0.05$, $\lambda_{\text{depth}} = 0.07$, $\lambda_{\text{param}} = 0.02$, and $\lambda_{\text{reg}} = 0.001$. All coefficients were chosen empirically to balance photometric fidelity, geometric consistency, and

physical plausibility. To ensure a fair comparison, we evaluated baseline methods using their official public implementations and recommended parameters wherever available.

Comparison Methods. For geometric reconstruction, we compare against implicit surface methods including UW-SDF* [CTW*24] and MonoSDF [YPN*22], explicit Gaussian-based methods including 2DGS and GOF [YLF*24], and traditional staged pipelines using PUIE-Net [FWH*22] enhancement followed by COLMAP [SZFP16], 2DGS, or GOF reconstruction. For image restoration, baselines include physics-based method IBLA [PC17], learning-based enhancement models PUIE-Net and FA+ [JYC*23], and Gaussian Splatting-based restoration technique SeaSplat [YLG24]. SeaSplat is a key baseline that also uses a physical model with Gaussian Splatting, though it is optimized for appearance restoration. To enable a comprehensive geometric comparison, we evaluate its geometry by first extracting the 3D Gaussian centers as a point cloud and then applying a standard Poisson Surface Reconstruction to generate meshes.

5.4. Geometric Reconstruction Results

5.4.1. Quantitative Evaluation

Synthetic Data (DTU-Water). Table 1 demonstrates WaterGS's superiority across synthetic underwater scenarios. Our method achieves the best Mean Chamfer Distance on four of five scenes, with an average performance of 1.18mm compared to 1.57mm for the best competitor—a 16% improvement.

Our analysis reveals that competing methods often must sacrifice geometric accuracy for completeness, or vice-versa. For instance, methods like GOF achieve high accuracy but poor completeness (scan40: 1.04mm vs. 2.33mm), while pre-enhanced approaches like 2DGS(PUIE) improve completeness at the cost of accuracy. WaterGS balances both: on scan40, competitive accuracy (1.01mm) with improved completeness (1.36mm) yields superior overall performance (1.18mm vs. 1.68mm for GOF). This validates that end-to-end, physically-based optimization outperforms staged paradigms by avoiding error propagation.

Real-World Data (UWBundle). Our advantages become more pronounced on real-world data (Table 2). WaterGS achieves 3.29 cm Mean error versus 3.70cm for the best baseline—a 12% improvement. Our F-score of 79.09% represents a substantial 12% relative improvement.

Notably, SfM-initialized baselines like MonoSDF and the GOF variants failed entirely on this real-world data, validating our robust VGGT initializer where traditional SfM proves too brittle. Among the methods that did produce a result, UW-SDF* suffered from high geometric error (8.68cm Mean) and poor completeness.

For a direct geometric comparison, we also evaluate SeaSplat (Tables 1-2). While it excels in appearance restoration, the use of 3D Gaussians results in significant geometric errors and noisy meshes with noticeable artifacts (Fig. 6), thereby validating our choice of a surface-based approach.

5.4.2. Qualitative Evaluation

Our qualitative results (Figures 4-7) support our quantitative analysis. On DTU-Water, GOF(PUIE) produces fragmented surfaces,

Table 1: Geometric accuracy evaluation on the DTU-Water dataset (unit: mm). The best, second, and third best are marked in red, yellow, and green, respectively. P+C+D refers to the PUIE+COLMAP+DENSE pipeline. UW-SDF* indicates our faithful re-implementation.

Method	scan40			scan55			scan65			scan118			scan122		
	acc↓	com↓	mean↓	acc↓	com↓	mean↓	acc↓	com↓	mean↓	acc↓	com↓	mean↓	acc↓	com↓	mean↓
P+C+D	1.1490	1.6533	1.4011	1.6696	0.5432	1.1064	1.4792	2.5640	2.0216	1.1042	2.0345	1.5693	1.7528	1.9146	1.8337
MonoSDF	3.2137	3.3971	3.3054	5.8533	6.8889	6.3711	4.5301	6.0910	5.3106	3.4356	4.0780	3.7568	3.3565	2.7248	3.0407
UW-SDF*	2.0591	1.5557	1.8074	2.9936	2.2183	2.6060	2.7196	2.7112	2.7154	1.8270	1.4282	1.6276	2.8313	2.0177	2.4245
SeaSplat	5.0970	4.5375	4.8173	3.6279	3.3198	3.4739	6.2871	4.9735	5.6303	2.5396	2.3794	2.4595	2.9994	2.5989	2.7992
2DGS	2.3240	2.0787	2.2013	2.1824	0.8987	1.5406	2.1805	1.3652	1.7728	2.3843	1.8343	2.1093	2.3128	1.6046	1.9587
2DGS (PUIE)	1.8622	1.0920	1.4771	2.4433	0.6349	1.5391	1.4627	1.2528	1.3578	2.3463	1.1511	1.7487	1.6943	1.7170	1.7056
GOF	1.0363	2.3281	1.6822	0.8399	1.1491	0.9945	0.8723	2.0839	1.4781	0.8128	4.7517	2.7823	1.3842	4.9290	3.1566
GOF (PUIE)	1.2801	1.9886	1.6344	1.1698	0.7106	0.9402	0.9978	1.8587	1.4283	0.7335	2.4950	1.6142	1.4285	3.0229	2.2257
Ours	1.0141	1.3628	1.1884	0.7565	1.2754	1.0159	0.9855	1.5466	1.2660	0.9237	1.5383	1.2310	1.5711	0.7016	1.1363

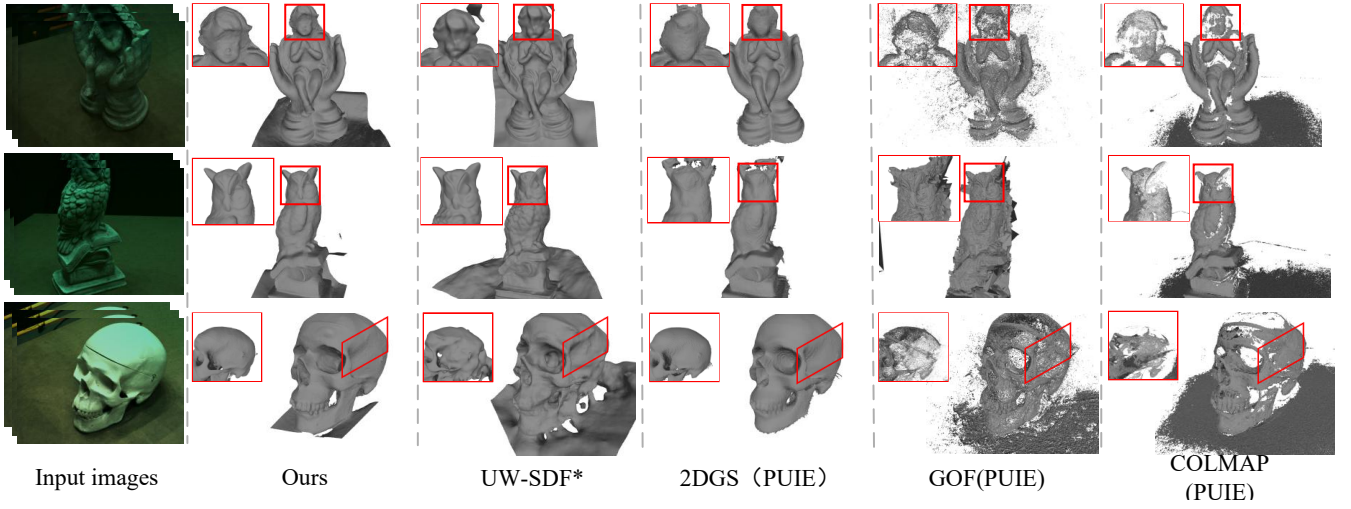


Figure 4: Qualitative comparison of geometric reconstruction results on synthetic DTU-Water dataset scenes. WaterGS (ours) demonstrates superior completeness and detail recovery compared to other representative methods.

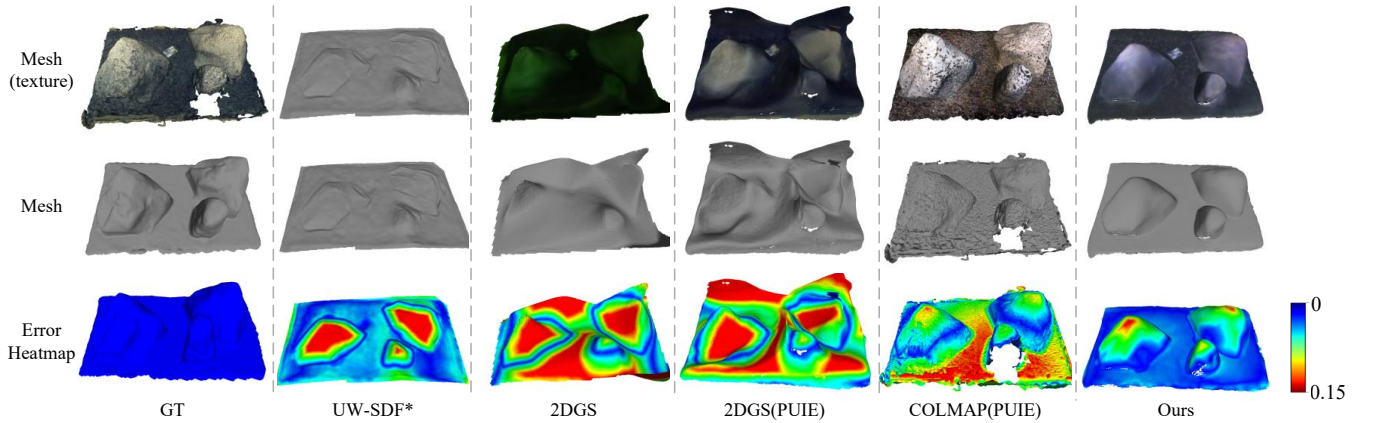


Figure 5: Qualitative geometric reconstruction results on the real-world UWBundle dataset.

Table 2: Geometric accuracy and reconstruction quality on the UWBundle real-world underwater dataset (unit: cm). ¹ indicates a method failed to produce valid output on this dataset. [†] indicates methods applied to PUIE-Net enhanced images. * indicates a faithful reproduction based on the original paper using default settings. V2DGS denotes the VGGT-initialized 2DGS pipeline. SeaThru denotes SeaThru-NeRF.

Method	Acc.↓	Comp.↓	Mean↓	Prec.↑	Rec.↑	F-score↑
COLMAP [†]	3.77	3.63	3.70	70.38%	70.71%	70.55%
MonoSDF			Failure ¹			
SeaThru	8.10	10.60	9.35	43.20%	56.90%	34.80%
NeurSea			Failure ¹			
UW-SDF*	5.13	12.23	8.68	70.50%	48.10%	57.19%
UW-SDF*	5.13	12.23	8.68	70.50%	48.10%	57.19%
SeaSplat	15.85	3.19	9.52	23.44%	84.45%	36.70%
2DGS	8.47	8.88	8.68	38.15%	33.24%	35.52%
2DGS [†]	9.69	8.67	9.18	32.46%	31.97%	32.21%
V2DGS [†]	8.49	6.50	7.50	44.50%	48.72%	46.51%
GOF			Failure ¹			
GOF [†]			Failure ¹			
Ours	2.58	3.99	3.29	85.55%	73.54%	79.09%

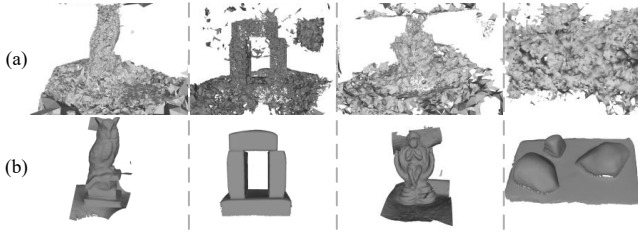


Figure 6: Geometric comparison with SeaSplat. Row (a) shows meshes generated from SeaSplat's point cloud via Poisson reconstruction, revealing significant noise and floating artifacts common to volumetric 3DGS representations. In contrast, our approach in row (b) produces clean and geometrically coherent surfaces.

and "PUIE+COLMAP+DENSE" (P+C+D) yields noisy reconstructions. In contrast, WaterGS generates a cleaner, more complete surface that preserves fine geometric details and fidelity.

This extends to real-world scenarios. On UWBundle (Fig. 5), our reconstruction demonstrates superior fidelity to ground truth. Fig. 7 showcases performance on in-the-wild scenes, where WaterGS reconstructs complex coral structures with fine detail, demonstrating practical utility for scientific applications.

5.5. Image Restoration Results

In addition to geometric accuracy, we evaluate the clear appearance J quality recovered by WaterGS across datasets with and without ground-truth references.

5.5.1. Quantitative Evaluation

Reference-Based Evaluation (UWBundle). The UWBundle dataset [SIJR17] provides in-air ground-truth images, enabling

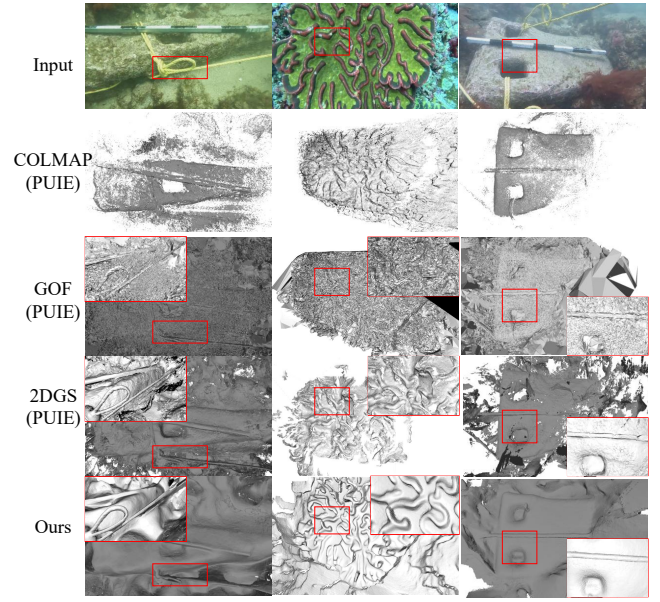


Figure 7: Qualitative geometric reconstruction results by WaterGS on challenging underwater scenes.

Table 3: Image restoration evaluation on the UWBundle real-world underwater dataset. Best and second-best values are highlighted.

Method	PSNR↑	SSIM↑	LPIPS↓
IBLA	17.532	0.8245	0.4503
SeaThru-NeRF	9.851	0.4943	0.7191
NeurSea	13.72	0.725	0.711
SeaSplat	16.710	0.8353	0.5152
FA+	16.396	0.8452	0.3129
PUIE	20.572	0.8992	0.1992
Ours	19.8861	0.9367	0.2939

full-reference evaluation using PSNR, SSIM [WBSS04], and LPIPS [ZIE*18] metrics (Table 3). Our results reveal a critical advantage of 3D-aware restoration. While PUIE-Net [FWH*22] achieves higher PSNR (20.57 vs 19.89), WaterGS obtains the best SSIM score (0.9367)—a 4.2% improvement over PUIE-Net (0.8992). This distinction is crucial: PSNR measures pixel-level differences, which can mislead in underwater restoration where color shifts occur, while SSIM measures structural fidelity. Our superior SSIM indicates WaterGS captures scene structure more accurately than 2D enhancement methods. As a joint framework optimizing both geometry and appearance, WaterGS produces restoration structurally superior to methods dedicated solely to image enhancement, maintaining geometric consistency through 3D scene understanding.

No-Reference Evaluation (DRUVA & Coral). On in-the-wild datasets such as DRUVA [VKR23] and our Coral dataset, we assess single-image quality and multi-view consistency using no-reference metrics (Table 4 & Fig. 9).

Multi-view Consistency: WaterGS achieves the best Mean

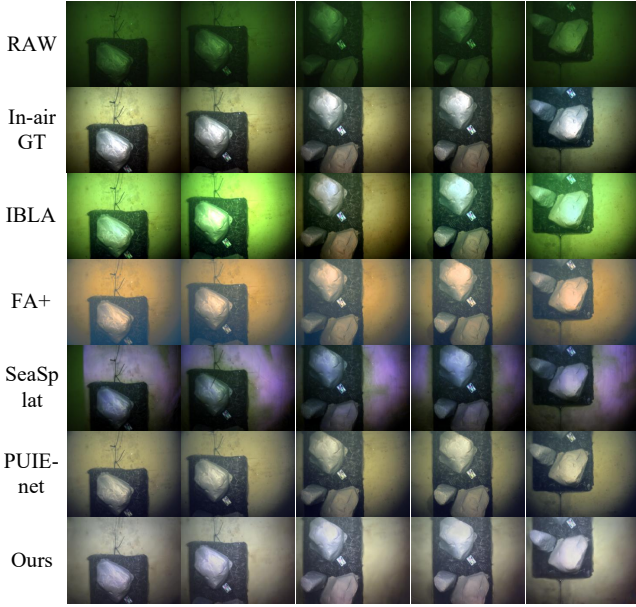


Figure 8: Comparison of restored images by WaterGS vs. In-air ground truth on the UWBundle dataset.

Table 4: Image restoration quality and multi-view consistency evaluation (no-reference). Best and second-best values are highlighted.

Method	NIQE↓	UCIQE↑	LPIPS Mean↓	LPIPS Std.↓	Hist Mean↑	Hist Std.↓
IBLA	3.6304	0.4964	0.4395	0.1115	0.8024	0.2495
SeaSplat	3.1206	0.4030	0.3906	0.1039	0.7816	0.3447
FA+	3.4034	0.3631	0.3978	0.0888	0.9293	0.1351
PUIE-Net	3.1990	0.4113	0.4033	0.1013	0.9251	0.1292
Ours	2.8491	0.3870	0.3801	0.0966	0.9168	0.1453

Pairwise LPIPS of 0.3801, a 2.7% improvement over SeaSplat [YLG24]. This evaluation highlights the importance of temporal consistency for underwater appearance restoration in 3D applications. While single-frame enhancement methods like FA+ and PUIE-Net may perform well on individual images, they cannot guarantee cross-view coherence. Our superior consistency validates WaterGS’s gaussian bundle adjustment scheme, which jointly refines poses, geometry, and appearance under physical constraints.

Single-Image Quality: WaterGS records the lowest NIQE [MSB12] score of 2.8491, a 9% improvement indicating superior naturalness and perceptual quality. For UCIQE [YS15], WaterGS achieves a competitive score (0.3870); while some traditional methods like IBLA [PC17] score higher on UCIQE, their restored images can appear perceptually unnatural, whereas WaterGS balances clarity with natural visual appearance.

5.5.2. Qualitative Evaluation

Visual results support our quantitative findings. Fig. 8 showcases sharp, structurally sound restoration on UWBundle, confirming the superior structural fidelity indicated by our SSIM scores. In con-

trast, the SeaSplat result in the same figure exhibits a strong violet color cast, highlighting a potential generalization issue of its physical model to the specific water conditions of this dataset. Fig. 9 and Table 4 demonstrates effectiveness across challenging scenes from DRUVA and Coral datasets, where WaterGS removes color cast and haze while restoring natural tones. Crucially, these restored sequences also maintain high visual consistency across viewpoints.

5.6. Analysis and Ablation Studies

To validate each key component of WaterGS and analyze our gaussian bundle adjustment scheme, we conducted comprehensive ablation studies and detailed component analyses.

5.6.1. Iterative Refinement via Gaussian Bundle Adjustment

To demonstrate our Gaussian Photometric Bundle Adjustment’s iterative refinement capability, we visualize reconstruction quality on the challenging DRUVA A3 scene at three stages: without pose optimization (w/o BA) using only initial VGGT poses, at intermediate convergence (Mid-BA, 30% iterations), and full convergence (Full-BA). Fig. 10 illustrates clear progression. The initial reconstruction (w/o BA) suffers from geometric artifacts with indistinct structures and blurred edges. Mid-BA shows considerable improvement with clearer object separation and coherent surfaces. The final Full-BA result displays highest geometric fidelity—structures are sharply defined, and fine details such as rope texture emerge. This demonstrates how our Gaussian bundle adjustment iteratively converges by simultaneously optimizing poses, geometry, and appearance under physical constraints.

Table 5: Ablation study on the UWBundle dataset (unit: cm). The best performing configuration (Ours - Full Model) is highlighted.

Method	Acc (cm)↓	Comp (cm)↓	Mean (cm)↓
w/o VGGT Init.	14.41	12.21	13.31
w/o Underwater Model (UM)	3.97	7.03	5.50
w/o Residual Term	5.39	8.42	6.91
w/o $\mathcal{L}_{\text{feedback}}$	4.28	7.03	5.66
w/o Gaussian BA (BA)	4.50	7.18	5.84
Ours (Full Model)	2.58	3.99	3.29

5.6.2. Component Ablation Studies

We conducted systematic ablation studies on UWBundle to quantify our core design choices: initialization strategy, physically-based underwater imaging model, and pose optimization module. Results are in Table 5 and Fig. 11.

Initialization Strategy Impact. Replacing our robust VGGT initializer with traditional COLMAP SfM (w/o VGGT Init.) caused a dramatic performance collapse, with the Mean error increasing 305% from 3.29cm to 13.31cm. This confirms that robust learning-based initialization is critical where standard SfM is brittle. COLMAP initialization often fails entirely (as with GOF methods in Table 2).

Physically-Based Imaging Model Necessity. Ablating our underwater imaging model (w/o UM) and optimizing against raw images

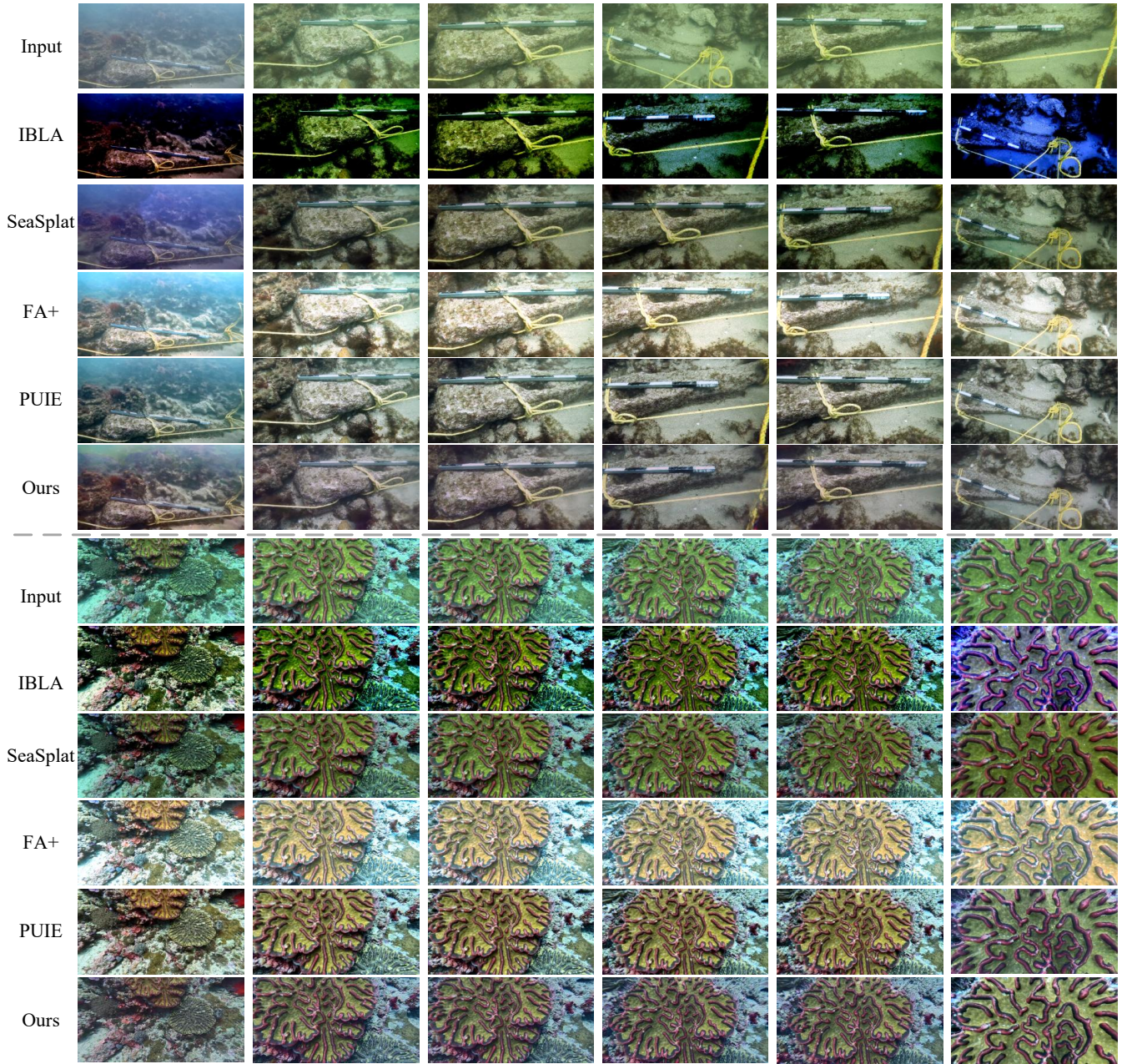


Figure 9: Qualitative comparison of image restoration by WaterGS and other methods on challenging DRUVA and Coral scenes. Our WaterGS method effectively removes color cast and haze, restoring natural colors and details with multi-view consistency.

led to a 67% error increase (to 5.50 cm). Fig. 11 shows a severely washed-out appearance with distorted geometry, proving explicit optical modeling is essential. Additional UM component ablations in Table 5 further increase *Mean* error to 6.91 cm (w/o Residual) and 5.66 cm (w/o $\mathcal{L}_{\text{feedback}}$).

Pose Optimization Criticality. Disabling our Gaussian Bundle Adjustment (w/o BA) substantially degraded performance, with

78% error increase (to 5.84cm). Qualitative results show blurry, ill-defined geometry, underscoring joint pose refinement’s criticality.

Component Importance Ranking. Robust initialization provides the largest improvement (COLMAP shows a 305% increase in Mean error vs. our full model), followed by Gaussian bundle adjustment (+78%) and the underwater imaging model (+67%), validating our integrated design.

6. Conclusions

This paper introduced WaterGS, a novel framework that addresses the significant challenges of underwater 3D reconstruction by formulating the task as an inverse rendering problem. Built on 2D Gaussian Splatting, our holistic approach integrates a learnable underwater optical model with robust pose refinement through Gaussian bundle adjustment that jointly optimizes scene geometry, intrinsic appearance, and camera poses, effectively disentangling complex underwater light transport effects from true scene properties. Comprehensive experiments demonstrate that WaterGS significantly outperforms existing methods, achieving substantial improvements in geometric accuracy while maintaining superior multi-view consistency—a critical advantage for 3D applications. Our framework successfully delivers both high-fidelity explicit 3D meshes and high-quality appearance restoration directly from raw underwater imagery, demonstrating the clear advantages of our unified approach over conventional staged pipelines and state-of-the-art alternatives.

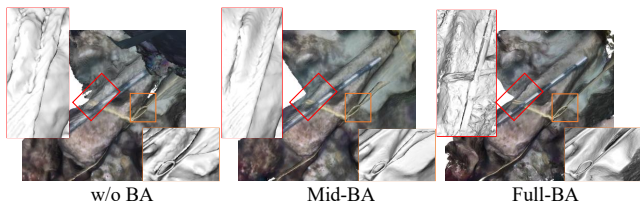


Figure 10: Qualitative mesh detail comparison on the DRUVA A3 scene, showing results for w/o BA (initial poses), Mid-BA (intermediate pose refinement), and Full-BA (final optimized poses) (Sec. 5.6.1).

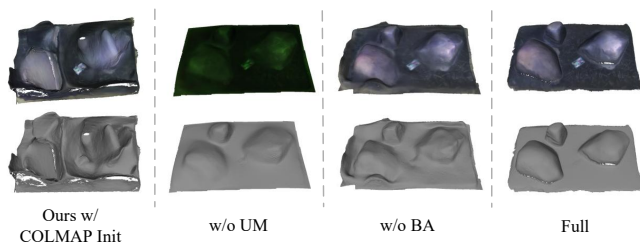


Figure 11: Qualitative ablation results (mesh & texture) on the UW-Bundle dataset, corresponding to Table 5.

7. Limitations and Future Work

WaterGS adopts a pinhole camera and first-order (single-path) scattering model; it therefore does not capture light-path bending from spatially varying refractive index (e.g., thermo/haloclines) or higher-order scattering and veiling glare under heavy turbidity or strong artificial lighting, which can bias pose/depth and reduce fidelity. The optimization also assumes a static scene, so moving particles, bubbles, or surface-caustic flicker may misalign multi-view supervision; and the joint optimization is compute-intensive and benefits from good initial poses and simple in-water calibration. Future work includes refraction-aware ray geometry, tractable

multi-scattering terms with source calibration, motion-aware optimization for dynamic content, and faster solvers or model distillation for large-scale reconstructions.

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